Report on waste bin collection

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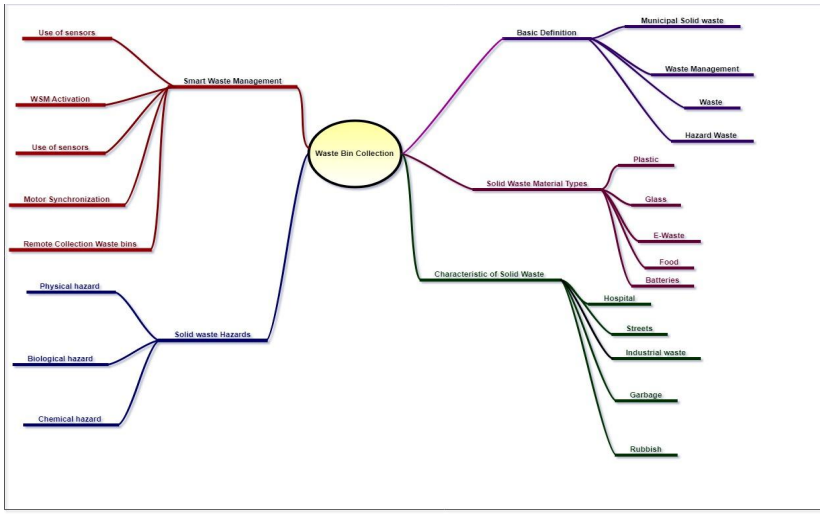

Figure 1: Mind map of Waste Bin collection System
(Source: Self-created using Draw.io)

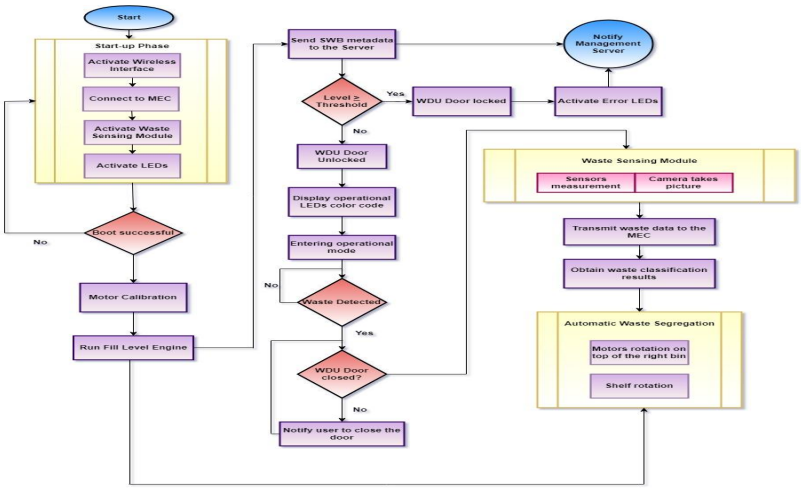

Figure 2: Flow diagram of Waste bin management
(Source: Self-created using Draw.io)

Discussion

The collection of the waste bins is an effective need and requirement for the better cleanliness of the society. The health of the community and the environment is very highly responsible for the cleanliness and the methods of collection of waste bins of the society. Effective measures and techniques are

needed to be adapted for the effective collection of the waste bins using smart waste bin management systems.

Current State

The waste bin collection should be sustainable, there are few companies like Bigbelly, Ecuba and few more which provides the proper and sustainable waste management solution and it is comprehensive, it can be tailored to various industrial and urban bin collection system and other companies also needs to learn and implement the same techniques to contribute in the sustainable process (Khan *et al.*2021). The society needs to adopt and cooperate with the garbage collector that will motivate them to do their task with greater responsibilities. The Government also needs to invest more money for the purpose and salary of the workers should be what they deserve.

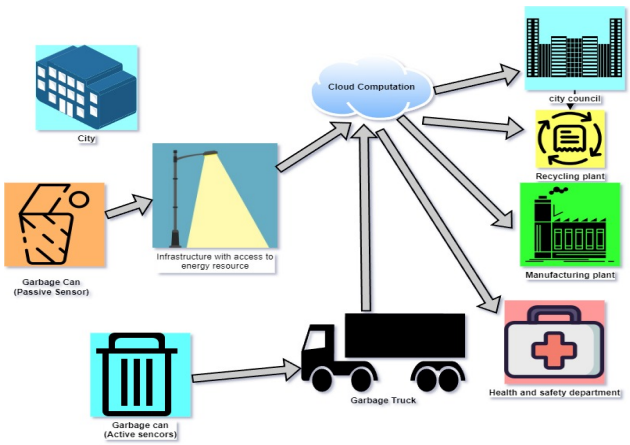

Figure 3: Cloud computation of waste bin management
(Source: Self-created using Draw.io)

Solution

For waste bin collection, there are some innovative solutions. In the figure which indicates the flow of the solution, there are some basic phases for waste bin collection. First step is the startup phase which executes checks for all sensors and devices. After all checks, SWB (Standard waste box) will be started (Pamintuan *et al.*2019). Then the LEDs are turned red while the flap door is locked. After start-up, the next step is Motors synchronization and calibration which controls the automatic waste and aligns the waste disposal unit. It will ensure the correct disposal of trash. For this step, a motor will run 360 degree full start-up spin using sensors. The next step is to use a fill level engine which is used to verify the SWB and see the storage area to store new pieces of trash after the controlling operations. When one bag fills the level of 100% capacity, the activation is interrupted and the door is locked (Harith *et al.*2020). This time, LEDs will also turn red, waiting to take action. The next step is waste insertion which is used when the SWB is active. Users can throw the trash to open the lid of the waste disposal unit.

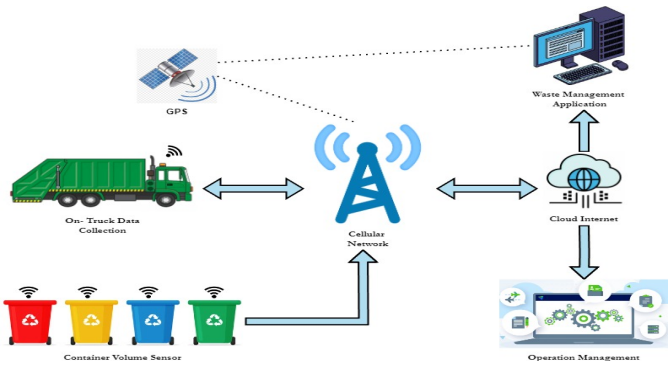

Figure 4: Architecture of smart waste bin collection
(Source: Self-created using Draw.io)

Challenges

The public is not aware of the need of proper waste management and are not showing as much interest as they should, the proper campaign regarding the issue is necessary also the modern technology is not involved in the waste management system, the society don't pay enough respect to the workers working for the said purpose and for that reason the workers are not motivated enough to do their job properly (Fatimah, Widianto & Hanafi, 2020). Despite the fact that the Government has banned plastics for commercial or for any use, plastic is being used regularly. Plastic is directly connected to the pollution as it is non-biodegradable and it doesn't decompose on its own and hence is the major source of the pollution.

Conclusion

Some innovative and complex solution will be required to address the issue of the waste bin collection process as well as the overall waste management. There are a lot of difficulties like limited space, low awareness of the residents regarding the bin collection process and making the urban waste management process more complicated. Involvement of the government in the waste collection process can improve the efficiency of the system, more funding will give a good kick to the process.

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